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RESEARCH-ARTICLE

Generating Real-Time, Selective, and Multimodal Haptic Effects from Sound for Gaming Experience Enhancement

GYEORE YUN, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

MINJAE MUN, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

JUNGEUN LEE, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

DONG-GEUN KIM, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

HONG Z TAN, Purdue University, West Lafayette, IN, United States

SEUNGMOON CHOI, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

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Published: 19 April 2023

Citation in BibTeX format

CHI '23: CHI Conference on Human Factors in Computing Systems
April 23 - 28, 2023
Hamburg, Germany

Conference Sponsors:
SIGCHI

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Gyeore Yun
ykre0827@postech.ac.kr
Pohang University of Science and
Technology
Pohang, Gyeongsangbuk-do
Republic of Korea

Minjae Mun
csmmj4594@postech.ac.kr
Pohang University of Science and
Technology
Pohang, Gyeongsangbuk-do
Republic of Korea

Jungeun Lee
jungeunlee@postech.ac.kr
Pohang University of Science and
Technology
Pohang, Gyeongsangbuk-do
Republic of Korea

Donggeun Kim
dgkim@postech.ac.kr
Pohang University of Science and
Technology
Pohang, Gyeongsangbuk-do
Republic of Korea

Hong Z. Tan
hongtan@google.com
Google LLC
Mountain View, California, United
States

Seungmoon Choi
choism@postech.ac.kr
Pohang University of Science and
Technology
Pohang, Gyeongsangbuk-do
Republic of Korea

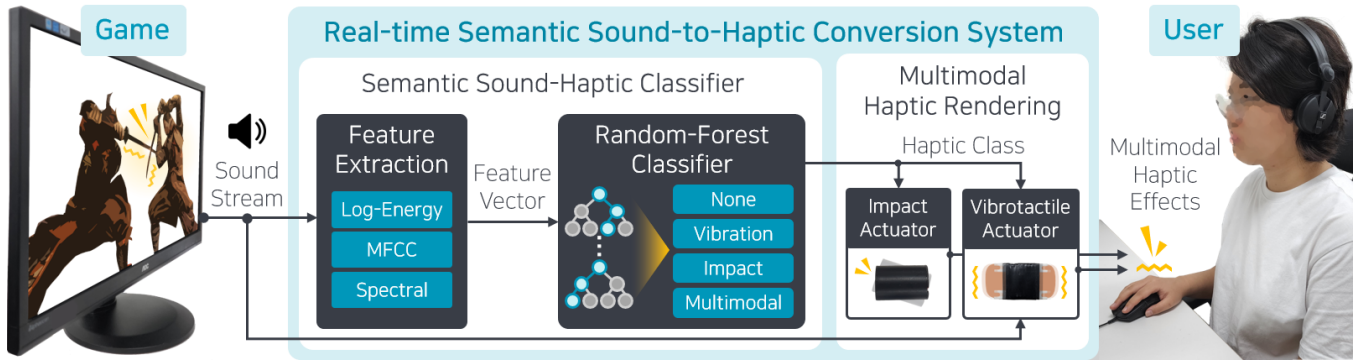


Figure 1: A user plays a game along with automatically generated multimodal haptic effects consisting of impact and vibrotactile stimuli in real time. At the moment a game sound suitable for providing a haptic effect (e.g., a gunshot or a sword clash) is produced, a device equipped with vibrotactile and impact actuators provides a multimodal haptic stimulus to the user.

ABSTRACT

We propose an algorithm that generates a vibration, an impact, or a vibration+impact haptic effect by processing a sound signal in real time. Our algorithm is selective in that it matches the most appropriate type of haptic effects to the sound using a machine-learning classifier (random forest) that is built on expert-labeled datasets. Our algorithm is tailored to enhance user experiences for video game play, and we present two examples for the RPG (role-playing game) and FPS (first-person shooter) genres. We demonstrate the effectiveness of our algorithm by a user study in comparison to other state-of-the-art (SOTA) methods for the same cross-modal conversion. Our system elicits better multisensory user experiences than the SOTA algorithms for both game genres.

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CHI '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9421-5/23/04.
<https://doi.org/10.1145/3544548.3580787>

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Applied computing → Computer games.

KEYWORDS

sound-haptic conversion, audio-haptic conversion, multimodal haptic effects, automatic generation, game

ACM Reference Format:

Gyeore Yun, Minjae Mun, Jungeun Lee, Donggeun Kim, Hong Z. Tan, and Seungmoon Choi. 2023. Generating Real-Time, Selective, and Multimodal Haptic Effects from Sound for Gaming Experience Enhancement. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 17 pages. <https://doi.org/10.1145/3544548.3580787>

1 INTRODUCTION

How fun would it be if we were the protagonists of video games who could vividly feel what our characters go through in the game? To afford such realistic experiences, real-time haptics is one of the most essential technologies [35, 39]. To this end, we can nurture

video game modelers and programmers to attain sufficient expertise in haptics, and then they can craft haptic effects for games to provide high-quality haptic effects whenever appropriate. This manual process may promise the best game experiences, but it is extremely costly both in time and money.¹

Alternatively, we can generate haptic effects *automatically* by processing the audiovisual output stream of games [38, 81]. Such an independent system, running in parallel with video game programs, is incomparably convenient and economical for game developers. However, it is difficult to assure the quality of such algorithm-based haptic feedback. Moreover, the conversion system must run in real time in the sense of crossmodal sensory synchrony. Haptic feedback significantly delayed from audiovisual events can feel unnatural and may lead to degraded user experiences.

The prevalent approach to automatically generating haptic effects for games is to capture the game's sound stream and convert it to a tactile stimulus produced by a vibration actuator. Physically, one event, e.g., a collision, makes both sound and vibration signals transmitted as acoustic and mechanical waves, respectively. The two signals of the same origin naturally have a high correlation, which supports that the sound-to-vibration conversion approach is fundamentally reasonable. It is also more efficient computationally than the other approach of image sequence-to-haptic conversion; see Section 2 for a review. This merit is critical for the real-time generation of haptic effects.

Early methods of sound-to-tactile conversion used relatively straightforward signal processing algorithms [41, 46]. Such algorithms can be adequate for some purposes, e.g., to enhance music listening experiences by providing additional haptic feedback. However, improving gaming experiences requires conversion algorithms that are substantially more sophisticated. This is because most games contain multiple soundtracks from different sources for a variety of purposes. Some of the game sounds, e.g., gunfire and engine sounds, are appropriate for haptic conversion. Other sounds, e.g., background music or human conversations, may not be adequate for haptification or can introduce masking vibrations that detract from the enjoyment of haptic effects due to (virtual) physical events. Consequently, we need to be *selective* in sound-to-haptic conversion for games. To this end, current state-of-the-art (SOTA) methods [35, 38] use a continuous mapping of psychoacoustic features extracted from the game sound to the relevant properties of a haptic stimulus; see Section 2.1 for a detailed review. Whereas such algorithms are fast, the sound selection process is relatively implicit and independent of the sound contents. Thus, these methods often suffer from triggering haptic effects for sounds that do not correspond to physical events or missing haptic effects for suitable sounds.

Another important recent advance in haptics is the use of *multimodal* haptic feedback, which refers to combining multiple haptic sensations, typically by mixing the output stimuli of different types of haptic actuators, to enrich the range of haptic expressions [69]. A good example is to combine a long vibration at a lower intensity with a strong and short impulse, generated by a vibrotactile

and impact actuator, respectively [47]. To apply such an advanced technique to sound-to-haptic conversion, we need to determine the type of haptic stimulus that is best matched to the current sound sample. This demands a more explicit selection scheme of game sounds, e.g., one using a machine learning (ML) approach.

In this paper, we present a sound-to-haptic conversion system designed to enhance gaming experiences, featuring (1) selection and explicit classification of the sound samples for conversion using an ML classifier, (2) support of multimodal haptic effects by combining vibration and impact stimuli, and (3) real-time performance satisfying the stringent requirement of synchronous sound-to-tactile perception. For (1), we developed a sound type classifier on expert-labeled data using random forest to achieve high accuracy and real-time performance. For (2), we designed and implemented a haptic rendering module that outputs a vibration, an impact stimulus, or both depending on the sound type. For (3), we optimized all steps of the algorithm and system design while considering the perceptual thresholds of temporal asynchrony from a preceding auditory stimulus to a following haptic stimulus (generally 20–75 ms [3, 45]). In addition, the efficacy of our system was examined by user experiments where participants compared the gaming experiences with haptic feedback for two types of video games, RPG (role-playing game) and FPS (first-person shooter). The results demonstrated that our selective and explicit sound-to-tactile conversion system significantly outperforms the previous methods in terms of user experience improvement.

Our contributions lie in (1) proposing the first real-time, ML-based, and explicitly-selective generation algorithm and system of multimodal haptic effects (vibration and impact) from the sound stream of gameplay, and (2) validating the user experience enhancement afforded by the approach.

2 RELATED WORK

Haptic effects are widely employed to enhance the user experience of audiovisual content, including video games [15]. Haptic effects that are well synchronized with the content can improve immersion and realism [11]. Recently, various actuators and mechanisms have been devised to deliver different types of haptic stimulus, such as vibration [67], impact [40, 67], tension [40, 49], pressure [66, 84], and changing shape [59]. However, manual authoring of haptic effects for such haptic devices requires considerable expertise in the haptic hardware and human perception, and the design process itself is iterative by nature and time-consuming [36]. Using effective authoring tools can improve productivity, but most of them focus on vibrotactile effects only [13, 23, 30, 34, 37, 54, 55, 61, 83]. Systems that automatically generate haptic effects according to audiovisual contents can be very useful. Algorithms capable of real-time conversion are particularly applicable to highly interactive applications such as video games.

Researchers have used image sequences as a source for generating haptic effects by tracking camera movement in first-person scenes [21, 56], estimating spatially salient regions [29, 82], detecting specific events of sports [53, 75, 76], finding the moments of gunshots [39], and selecting scenes suitable for vibrotactile stimuli [9]. However, visual-to-haptic conversion still works for very

¹One example documented in the literature is for 4D films, which combines other multisensory stimuli, such as motion, vibration, and wind, with the audiovisual content. Typically, it takes approximately 16 days for three designers to produce the 4D effects for one film [36].

limited cases, largely because of the complexity of visual information and the lack of real-time algorithms.

In general, sound-to-haptic conversion is technically more feasible than visual-to-haptic conversion because of the smaller size of sound data and the similar properties between sound and vibrotactile signals. As such, there are more research studies and commercial products in this space, as reviewed in detail in Section 2.1. In particular, sound-based methods can be advantageous to video games where games in the same genre share similar sound characteristics. This property also facilitates scalability, e.g., the sounds of gunshots or punches in different games are more similar than their corresponding visual scenes.

2.1 Sound-to-Haptic Conversion

Early sound-to-haptic conversion systems employed signal processing algorithms (mostly filtering) to express the characteristics of a sound signal within the haptic bandwidth. This approach is the most efficient way with the least computational delay and is still frequently used for music-related applications. For example, Karam et al. mapped the sound energies in multiple frequency bands to multiple body locations with vibrotactile stimuli using a band-pass filterbank [25, 26]. However, the high-frequency features of sounds cannot be transformed into haptic effects using a simple band-pass filter technique. To convey the high-frequency components of music as haptic effects, researchers expressed the signal strength of the treble band using frequency shifting [8, 46] or a dual-band linear resonant actuator [24]. In addition, Seo et al. [57] mitigated the gap in perceived intensity between auditory gunshots and their haptic stimuli using the actuator's transfer function. Due to their simplicity, the signal processing-based sound-to-haptic conversion methods are also used in the industry. As a consumer product, the bHaptics vest uses sound energies in multiple frequency bands to actuate a large number of vibration motors on the body [6]. Headsets that emphasize low-pitch sounds with vibrations are commonly available for gameplay [10].

Selective sound-to-haptic conversion systems analyze sounds and present haptic effects only for specific targets, such as gunshots in games or beats in music. An excellent example is Astro's playroom [50], a game that shows off Sony Playstation's Dualsense controller. It provides well-crafted haptic effects only to special sound effects, but not to background music (BGM). Although these effects are pre-programmed, they nonetheless demonstrate the ideal use case of selective sound-to-haptic conversion.

The concept of automatic and selective sound-to-haptic conversion was formalized in Lee and Choi [35]. The authors proposed to design a simple mapping from two psychoacoustic features, loudness and roughness, computed from a sound signal to the perceived intensity and roughness of a vibrotactile signal. They then presented a method to synthesize a vibration stimulus with a specific perceived intensity and roughness. The mapping controls the type of sound for haptic conversion based on the sound's perceptual quality. For example, a mapping that emphasizes only perceptually loud and rough sound can render gunshots but filters out BGM, making the conversion selective (albeit implicitly). This approach does not rely on machine learning, so no needs to build datasets. It is simple and fast, and well suited for real-time execution. This

method was recently extended by Li et al. [38], where additional psychoacoustic variables (sharpness and booming) and a low-band energy were also used in combination with simple thresholding to detect sounds for selective haptic conversion. However, the classification performance of these methods can be inherently lower than the desired level. For example, the false positive rate reported by Lee and Choi [35] was up to 20%. The algorithms can also easily miss target sounds with low energy or generate unwanted haptic effects for loud and rough human voices.

These problems led us to adopt machine learning techniques for sound-to-haptic conversion. We are aware of two sound-based machine learning algorithms for the automatic generation of physical (haptic and motion) effects. Yoshida et al. [76] proposed an ML-based system that automatically generates vibrotactile signals based on the sound data from a recorded tennis video. Yun et al. [81] designed an ML-based system that detects the gunfire sounds of a game character using a deep neural network (DNN) and presents a motion effect on a motion platform to enhance 4D experiences. The two studies are limited by the use of clean and noiseless sounds only for training [76] and one character's firing sounds [81], respectively, and by the slow computation that necessitated off-line processing.

In summary, despite a plethora of studies and commercial products, there is still a need for sophisticated, fast, and selective sound-to-haptic conversion algorithms that can deal with various types of sounds and haptic stimuli that are tailored to the application domain.

2.2 Multimodal Haptic Rendering

In multimodal haptics, haptic stimuli generated by different types of actuators are superimposed to deliver richer haptic sensations [47]. Despite the popularity of vibrotactile feedback, it cannot reproduce other haptic modalities such as impact, pressure, shear force, and stress from physical interactions. As reviewed by Wang et al. [69], a large amount of past research can be attributed to multimodal haptics, and this novel research space is gaining more interest. Researchers have attempted to integrate vibrotactile stimuli with other haptic modalities, such as impact [31, 47, 52, 71, 72], kinesthetic force feedback [12, 42, 48, 60, 73], and thermal feedback [42]. There exist even handheld [65, 70] and wearable devices [74] that render pressure, shear force, pseudo force, shape, and texture, as well as commercially-available actuators and controllers emulating multiple haptic modalities, such as the impact actuators from Nanopoint Technology [44] and Acturonix [2] and the force feedback triggers in the Sony Dualsense controller [51].

Among the many, impact, a strong impulse of force applied to the surface in a very short time, is a very simple and effective modality in diversifying haptic sensations. Adding an impact actuator to a haptic device is also straightforward. Multimodal feedback of impact and vibration was demonstrated to improve the user experiences for collision events [47]. It is expected to enhance gameplay experiences as collisions and explosions are crucial events for haptic feedback in many games.

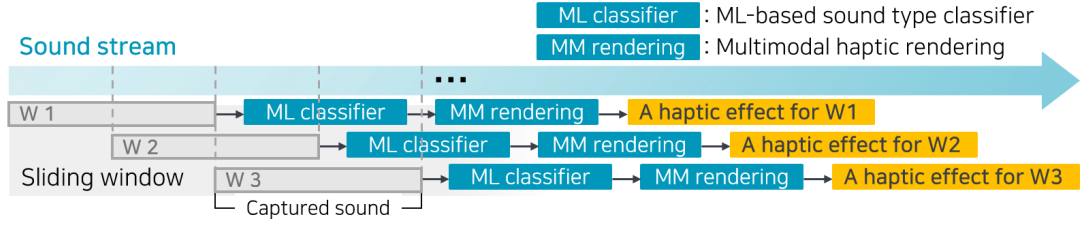


Figure 2: Structure of computational modules using sliding windows to generate a continuous haptic signal in real time.

3 SYSTEM OVERVIEW

We choose a use scenario where a user plays a video game while enjoying multisensory feedback of sound and haptic effects. The haptic effects are generated on the fly from the sound effects by an independent process. Additionally, the haptic effects can be multimodal, combining vibration and impact stimuli.

Our ML-based audio-to-tactile conversion with explicit selectivity consists of three major computational steps:

- (1) **Sound sample capture:** The system captures the latest sound sample of a certain length.
- (2) **Sound type classification:** The system selects the most appropriate type of haptic stimulus for the captured sound sample using an ML-based classifier.
- (3) **Haptic stimulus generation:** The system generates the haptic effects as selected by the classifier.

These three steps are repeated at a fast update rate to compose a continuous haptic stimulus. For this purpose, we adopt the sliding window structure, as shown in Fig. 2. The system captures a sound sample of the given window length, processes it, and selects a haptic signal for the sound sample.²The procedure is repeated with a delay that ensures that two consecutive sliding windows overlap for continuity in the haptic output.

A critical computation requirement here is that the sound and haptic stimuli be perceived as simultaneous in order for them to be attributable to the same physical events. However, a haptic stimulus always follows the target sound stimulus in audio-to-tactile conversion. Thus, we must minimize the computational load while improving the crossmodal conversion quality—a challenging computational-perceptual optimization problem.

Fig. 3 illustrates the timing diagram. Our system runs in parallel with a game program and accesses its audio output stream. Assuming that a sound starts at time 0 and its amplitude increases and reaches the first prominent peak at t_{peak}^{sound} . This sound signal is sent to the speaker, and it makes the physical sound with the first peak at $\tilde{t}_{peak}^{sound}$. This time is delayed by the inevitable sound latency caused by computation, communication, and actuation. Likewise, if a haptic signal is computed to have the first peak at F, the physical haptic stimulus generated from the haptic actuator has the first peak at $\tilde{t}_{peak}^{haptic}$ after the haptic latency. Then, the sound-to-haptic

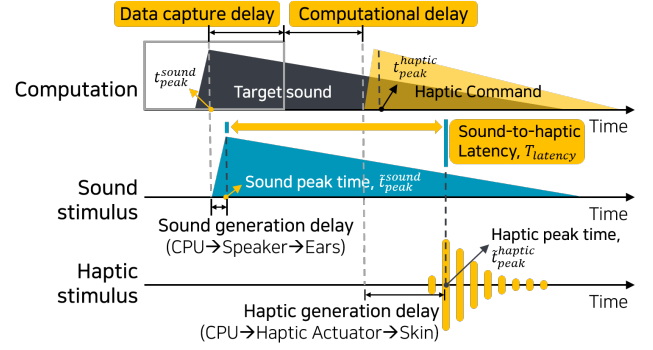


Figure 3: Timing diagram for sound-to-haptic conversion.

latency is determined as:

$$T_{latency} = \tilde{t}_{peak}^{haptic} - \tilde{t}_{peak}^{sound}. \quad (1)$$

This crossmodal sensory latency $T_{latency}$ must be less than or comparable to the temporal difference thresholds between sound and tactile stimuli, which range from 25 ms to 75 ms [3, 45].³

The sound and haptic latencies depend on the specific hardware used in the system and cannot be altered. What we can optimize is the computation time to convert a sound sample to a haptic stimulus, which consists of two intervals in our system. First, from the sound stream, we capture a sound sequence of the given window length. A longer sequence is likely to contain more information for sound type classification, but it also increases the delay. The window length determines the upper bound of this *data capture delay*. Thus, it is one critical parameter for optimization. Second, from the captured sound sample, we classify the sound type using an ML-based classifier and compute an appropriate haptic signal. This step also takes considerable time, incurring a *computational delay* for optimization.

All optimizations and empirical measurements reported in the following sections were made using the following hardware configuration. A commodity PC (Intel i9-12900F CPU, 32 GB RAM, on-board sound card, and Nvidia GeForce RTX 3080 graphics card) was used. The haptic device integrates a vibration and an impact actuator inside a commercial mouse, as shown in Fig. 4. We chose

²Sound is generally sampled at 44.1 kHz. The sampling rate for haptic signal reconstruction depends on the type of haptic stimulus. For vibration stimuli, the sampling rate required for faithful and smooth reconstruction is generally between 5 and 10 kHz.

³Strictly speaking, we should use the point at which the stimulus intensity first exceeds the detection threshold of the stimulus' modality to define the crossmodal sensory latency. Our definition in Eq. (1) is approximate to avoid the practical difficulty of measuring the precise detection thresholds that depend on many other conditions.

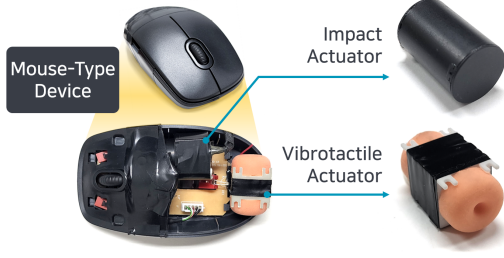


Figure 4: A mouse customized with impact and vibration actuators.

the mouse form factor as it is familiar to general users. The vibration actuator was a voice-coil actuator (Tactuator BM1C, Tactile-Labs [64]) that could generate vibrations in a wide frequency band of 30–1000 Hz (maximum intensity: 7.5 G at 55 Hz). The impact actuator (Digital Haptic Actuator, Acturonix [2]) could generate impact stimuli repeatedly up to 500 ips (impacts per second). Its maximum intensity was empirically measured to be 29.1 G (one full impact without repetitions) on our device.

4 SOUND TYPE CLASSIFICATION USING MACHINE LEARNING

For selective sound-to-haptic conversion, our system first determines whether the captured sound sample is suitable for haptic feedback. This task can be done in one of the two approaches:

- (1) Classify the semantic class of the captured sound, e.g., gunshot, car collision, and engine. Then designate the most appropriate haptic rendering method based on the sound class.
- (2) Map the most appropriate type of haptic stimulus or rendering method directly from the sound sample.

The first approach is general and hence potentially more powerful as each sound class can exploit a tailored haptic rendering method. However, we need to build a well-defined set of sound classes and even their ontology. The required amount of data and the resulting classifier can also be more complex. In the second approach, each sound sample is directly mapped to the most appropriate type of haptic feedback. For its expected simplicity, we selected this approach for four haptic types: none, vibration, impact, and multimodal (vibration+impact). A robust supervised classifier designed and implemented for this purpose is described in this section.

4.1 Characteristics of Target Sounds

From our daily interactions with physical objects around us, we expect haptic effects when our game character attacks with a weapon, a vehicle collides with an object, an explosion occurs, etc. Such events produce the target sounds of our system. Most of them are impulsive sounds which generally comprise three stages: the attack stage, the steady stage with the maximum loudness, and the release stage [16], as shown in Fig. 5. In the attack stage, the sound intensity (loudness) increases sharply within a short time interval. The attack stage should ideally be aligned in time with the beginning of the corresponding haptic feedback. The peak loudness is

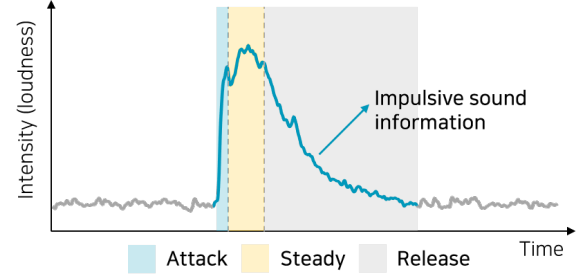


Figure 5: General loudness curve of impulsive sounds.

sustained during the steady stage. This is followed by the release stage in which the loudness gradually decreases, then followed by a reverberation.

4.2 Datasets

A supervised-learning classifier requires a dataset of input and output variables. For the input, we use auditory features rather than a raw sound signal to reduce the dimension and simplify the classification model. The output is the type of haptic effect for the input sound: none (NONE), impact (IMP), vibration (VIB), and multimodal (MM; simultaneous impact and vibration stimuli).

4.2.1 Data collection. We constructed datasets for the two video game genres of RPG and FPS, respectively. We collected sound samples from YouTube gameplay videos of four RPG games (Assassin's Creed 3 [68], Elden Ring [43], God of War: Ragnarök [63], and Sekiro [62]) and eight FPS games (Apex Legends [4], Battlefield 5 [5], PUBG Battlegrounds [32], Call of Duty: Black Ops Cold War [1], Fortnite [18], GTA 5 [19], Overwatch [17], and Valorant [20]). All of these games are popular around the world.

For each video, we calculated the loudness [85] of its audio stream. We then extracted sound samples by iterating the following procedure (Fig. 6): (1) Find the maximum loudness point in the first 1-s sound signal as a candidate for the beginning of the steady stage,

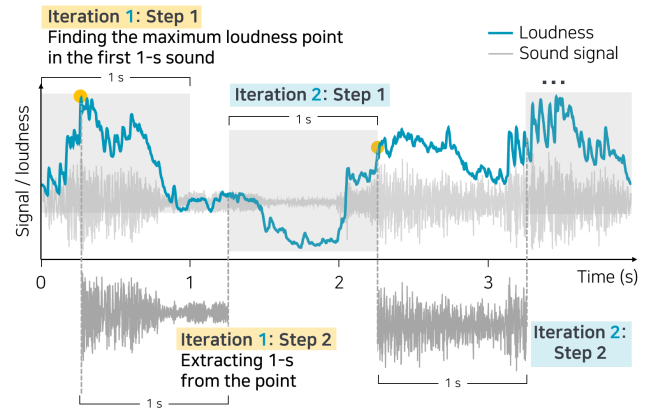


Figure 6: Iterative data collection process from the sound stream.

Table 1: Distribution of haptic types in the two datasets.

Class	RPG dataset	FPS dataset
NONE	660 (71.4%)	666 (68.9%)
VIB	42 (4.5%)	85 (8.7%)
IMP	77 (8.3%)	43 (4.4%)
MM	146 (15.8%)	187 (19.1%)

(2) Extract the 1-s sound sample starting from the maximum loudness point, and (3) Repeat (1) and (2) from the end of the extracted 1-s sound sample. The length of 1 s was chosen to be sufficiently long so that people can understand the sound samples' contexts when hearing them for labeling (we use much shorter sound samples for the actual ML classification; see Section 4.2.3). As a result, we collected 2461 sound samples for RPG and 7362 samples for FPS. Then, we randomly selected 1000 samples for each genre for the next labeling step.

4.2.2 Labeling. Three haptic experts participated in building datasets for the two game genres.⁴ They had experiences of using impact and vibration effects over at least one year. Each expert chose the type of haptic effect appropriate for each 1-s sound sample while focusing on the first sound event in the sample (a sound sample may contain multiple sound events for 1 s). The label of each sound sample was determined by the majority vote of the labels chosen by the three experts. The data samples labeled differently by all the experts were excluded. The labeling results by the three experts showed an agreement rate of 69.2% for RPG and 73.4% for FPS. Consequently, we obtained 925 labeled sound samples for RPG and 981 for FPS. Table 1 shows the distributions of the haptic effect types in the labeled datasets.

4.2.3 Data Processing and Augmentation. The 1-s sound samples used for labeling often contain multiple meaningful sound events; e.g., see the left plot in Fig. 7a. During labeling, the haptic experts concentrated on those first sound events, and we need to extract them to build a dataset for classifier training. Recall that we use sliding windows for data capture and haptic rendering, as explained in Section 3 and Fig. 2. Therefore, we use the same window size to extract the first sound events.

The window size has effects on both the data capture delay (Fig. 3) and the classification accuracy, as illustrated in Fig. 7b. A longer window increases the data capture delay because the sound event can only be detected after all the sound values in the window are captured. A shorter window decrease the data capture delay, but the captured data may not contain sufficient information for accurate and robust classification. Considering this trade-off, we searched for an optimal window size experimentally, as described in Section 4.4.

Likewise, the position in the window at which the sound has the maximum loudness (Fig. 7c) matters. If the maximum loudness position (i.e., the beginning of the 1-s sound sample) is close to the

beginning of the window, the amount of the target sound information increases, but the data capture delay also increases. This is similar to the effect of increasing the window size. If the maximum loudness position is close to the end of the window, we have the opposite effect. To balance the trade-off, we extract a first event sound such that its peak loudness is located at the center of the window (Fig. 7a).

We also adopt probabilistic audio data augmentation to improve the classification performance and robustness for diverse sound input data. Volume gain adjustment and white Gaussian noise addition are applied independently with an 80% probability. If applied, the volume gain is adjusted between -5 and 5 dB, and the white noise is added in 15–25 dB signal-to-noise ratio (SNR), both following a uniform distribution. We increased the data size ten times by repeating the probabilistic augmentation and finally obtained approximately 10,000 samples per dataset.

4.2.4 Feature Extraction. In general, machine learning models dealing with sound data exploit various sound features as input variables instead of using the raw signals directly to reduce the input dimensions and improve robustness. We extract an 18-dimensional (D) vector consisting of widely-used sound features: log energy, 13-D Mel-Frequency Cepstral Coefficients (MFCCs), and 4 spectral features of spectral entropy, spectral roll-off point, spectral slope, and harmonicity ratio [58]. Further, we divide the data capture window into three sliding sub-windows, as shown in Fig. 8, and extract the 18-D feature vectors from each sub-window. This is to reflect the temporal changes in the feature values. The sub-windows' length is $\frac{1}{2}$ of the data capture window length, and the sliding step is $\frac{1}{4}$ of the window length. Finally, we concatenate the 18-D feature vectors from the three sub-windows into one 54-D feature vector.

The 54-D feature vectors were calculated for all the sound samples collected. They were used as the input to build a sound type classifier described in Section 4.3. We also tested adding psychoacoustic features, such as loudness, roughness, and sharpness, as in the SOTA algorithms [35, 38]. However, it did not lead to improved classification performance. We did not use the psychoacoustic features in our system considering their additional computation cost.

4.3 Classification Model

To build a sound-type classifier, we used random forest (RF) [7], an effective machine learning model composed of an ensemble of decision trees.⁵ Random forest prediction is also computationally lighter than those based on deep learning. It avoids overfitting problems by using randomly-sampled training data (Bootstrap aggregating; Bag) and increasing the number of decision trees (Number of Bags; NoB).

⁴We also recruited seven regular users who were not haptic experts and conducted data labeling. However, their labeling results were quite inconsistent, e.g., inter-participant correlations around 0.3, and we stopped labeling efforts in this direction.

⁵We also tested other classifier models, including multi-layer perceptron (MLP; one of the simplest deep learning models) and support vector machine (SVM; another classical ML model). The MLP model was very successful in classification accuracy (higher than 99%), but its computational speed was too slow for our purpose (35 ms on average to classify one window data). The SVM model showed a false alarm rate of about 20%.

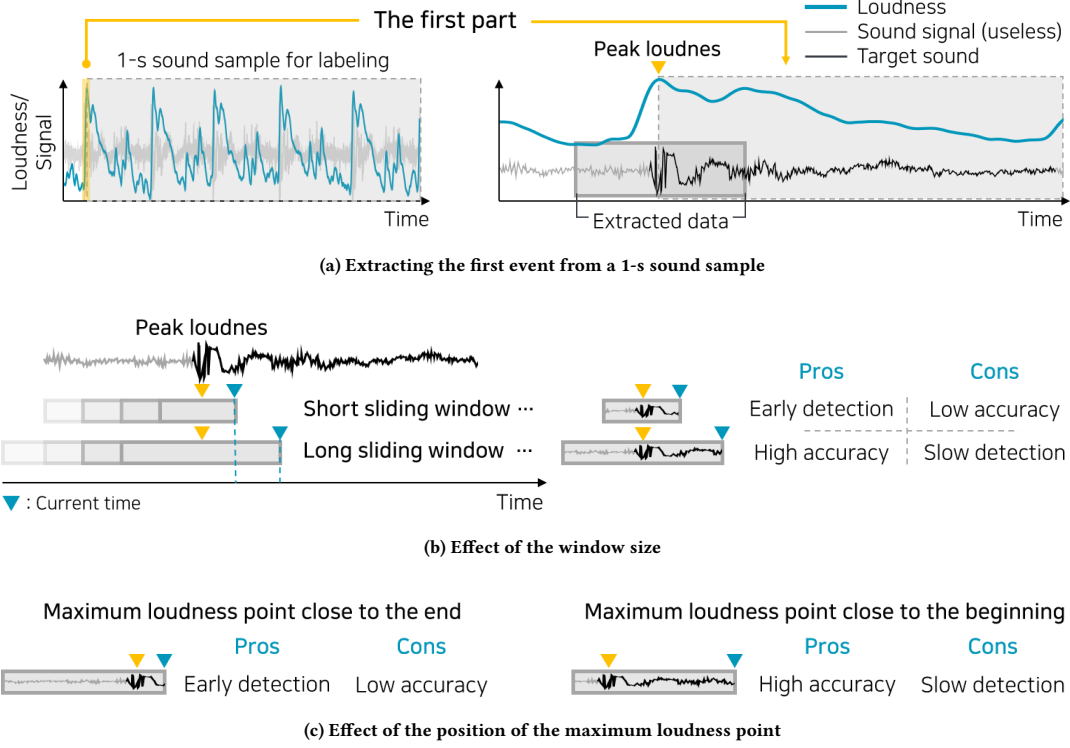


Figure 7: Sound data preprocessing.

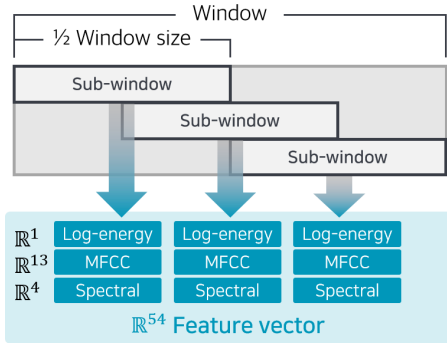


Figure 8: Extraction of 54-D feature vector from a window.

We built two respective RF classifiers for the two datasets of different game genres using a public Matlab function, TreeBagger, with NoB as a parameter. Increasing NoB is expected to improve the classification performance but also lengthens the computation time. To construct decision trees, the training data were sampled using randomization with replacement. A priori probability for each class was taken from the empirical frequencies of the classes in the individual datasets. The other parameters were set by their default values.

4.4 Experimental Results

Our sound-type classifier design has two key parameters, window size for capturing sound samples and NoB for training a random forest model. The selection of each value is a trade-off between classification accuracy and sound-to-tactile sensory latency. Therefore, we experimentally searched for the optimal values of window size and NoB.

Fig. 9 illustrates the effects of window size and NoB on the OOB error of each RF model. As expected, increasing the window size and the NoB improves the classification accuracy, but the improvement is saturated when the two parameters are sufficiently large. It is also experimentally confirmed that the computation time for classification increases almost linearly with the NOB, as shown in Fig. 10.

Based on these results, we choose the optimal values of window size = 40 ms and NoB = 60. These are the smallest values of window size and NoB, respectively, immediately after the OOB error saturates, as shown in Fig. 9. These optimal values support high performance levels. First, the OOB error is 0.013, which corresponds to the classification accuracy of 98.7%. Second, the average data capture delay is 20 ms, and the classification computation delay is approximately 2.5 ms (Fig. 10). Thus, the total computational delay is 22.5 ms on average. Adding the worst-case actuator delay (impact, 40 ms) and nothing that the computation time of haptic commands is almost instant and negligible, the average sound-to-haptic sensory latency is bounded by 62.5 ms. We can expect a shorter latency

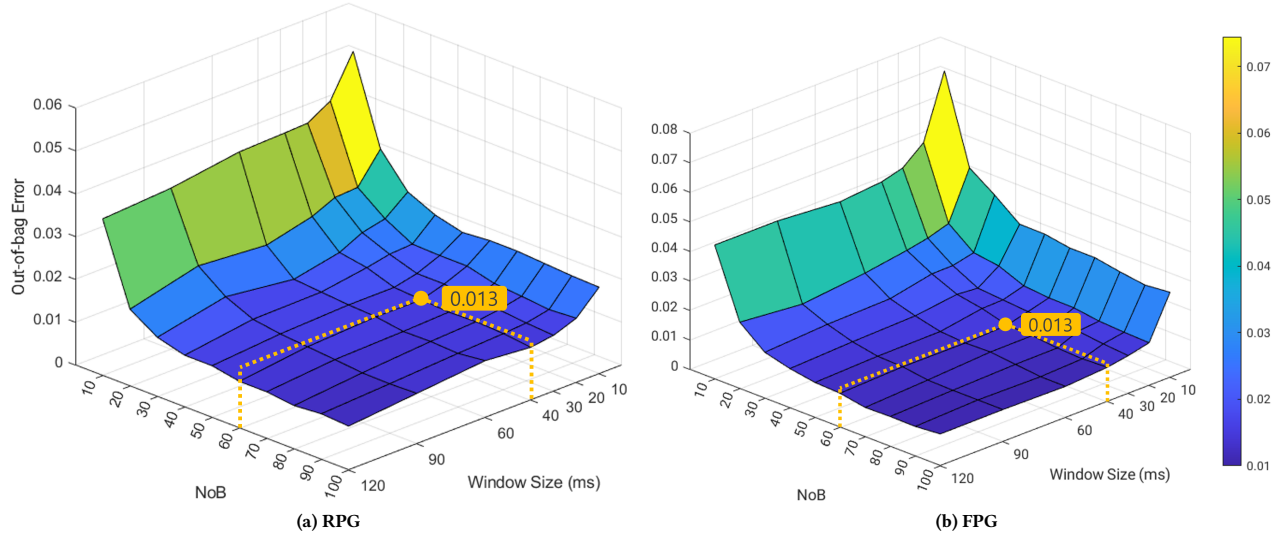


Figure 9: Out-of-bag errors as a function of window size and NoB. The yellow points (window size=40 ms and NoB=60) indicate the selected optimal parameters.

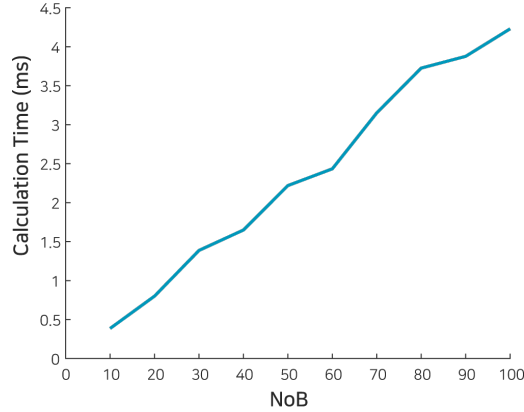


Figure 10: Average computation time for sound-type classification for different values of NoB.

for vibrotactile feedback as vibration actuators have much faster responses. This is well within the human sound-tactile crossmodal JND of asynchrony perception (20–75 ms [3, 45]).

Finally, Fig. 11 shows the confusion matrix of sound-type classification. Between NONE and the rest (the classes that require haptic effects), the true positive and false positive rates were 96.8% and 0.2% for RPG and 97.3% and 0.2% for FPS. These are substantially better scores than those reported by the previous methods; e.g., 20% of false positive in [35]. Also, it is noticeable that the VIB-to-NONE errors are 8.8% for RPG and 5.3% for FPS, and they are relatively higher than the other errors. This indicates that the datasets contain many NONE samples which sound like VIB. The sounds of vehicle engines or reverberations are representative examples of VIB. Without auditory contexts, sometimes they sound similar

to noise or environmental sounds of which the sound types are normally labeled as NONE. Such a phenomenon may have led our classifier to identify VIB as NONE.

5 MULTIMODAL HAPTIC RENDERING

After the random forest classifier determined the sound type for a sound sample, the next step is to compute an appropriate haptic signal, either vibration, impact, or vibration+impact, and send it to the haptic actuators. This procedure is outlined in Fig. 12 and explained in detail in this section.

5.1 Rendering Method

We provide a vibrotactile stimulus when a captured sound sample is classified as VIB or MM. The vibration signal is computed by applying a low-pass filter to the audio stream. This method is one of the most efficient methods with almost no delay. It also expresses the sound characteristics within the bandwidth of the vibration actuator with minimal distortions. The low-pass filter has a cut-off frequency of 250 Hz, at which frequency humans are most sensitive to vibrotactile stimuli [22]. The filtered sound stream is multiplied by a tactile volume gain, and its result is sent to the vibration actuator. The tactile volume is either 1 or 0, which merely turns on or off the vibration channel. This design guarantees the continuity of the vibration command signal.

An impact stimulus is generated when the sound class is IMP or MM. The impact actuator we use is a variant of a DC motor with a large rotating mass. If the mass hits a mechanical stop in the actuator, it makes a strong impulse delivered through the actuator case. The actuator's internal mechanism does not allow us to modulate the impulse intensity. Thus, the impulses were always delivered at the maximum intensity.

The impact actuator has a relatively large latency (40 ms) for full operation. This latency imposes a constraint on the rendering

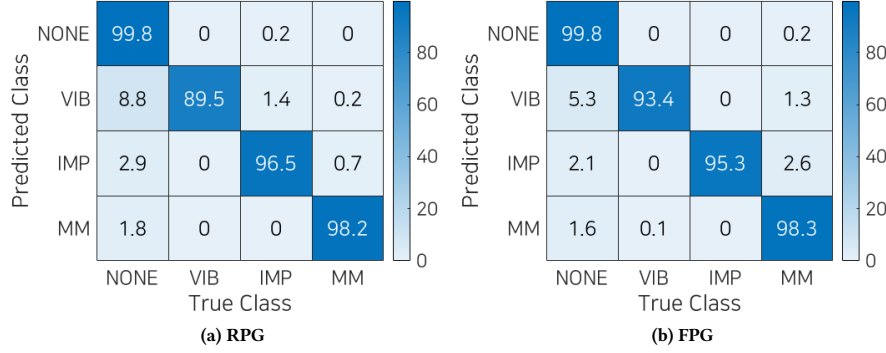


Figure 11: Confusion matrices of sound-type classification. Values in the cells are in percentages.

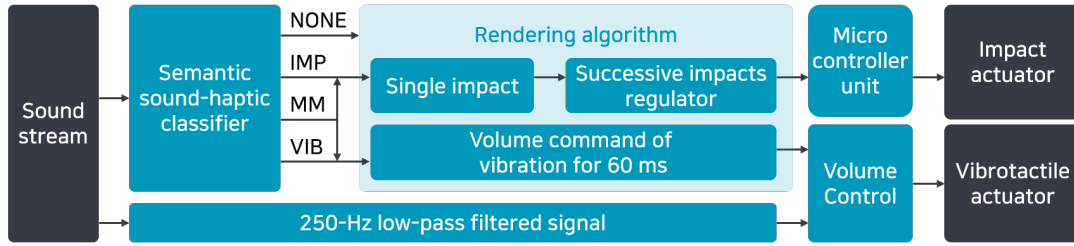


Figure 12: Process of ML-based selective sound-to-haptic conversion using multimodal haptic stimuli.

algorithm design. If two impulse commands are sent within a time interval shorter than the actuator latency, the actuator may not respond to the second command. Hence, we disable the impact actuator commanding for 40 ms after the first command.

5.2 System Implementation

The entire algorithms and processes from audio stream capture to haptic command generation are integrated into a program running on Microsoft Windows 10. The program was developed using Visual Studio C++ 2019. It captures the audio stream using the Windows Audio Session API (WASAPI), which enables accessing the audio data flow managed between the application and an audio endpoint device. For sound feature extraction and classification, we integrated C++ functions migrated using MATLAB Coder from the public MATLAB 2021b functions (mfcc, audioFeatureExtractor, TreeBagger). Parallelization by multithreading supported in MATLAB Coder is applied to feature extraction functions to accelerate computation speed.

For vibration rendering, we opened an additional audio channel using an audio processing object (APO). The average of the two stereo sound channels is copied into the new audio channel, which is then sent to the vibration actuator. It also allows tactile volume gain adjustment independently. The impact commands are transferred via serial communication to a microcontroller board (Arduino Due), which converts them to an analog signal to drive the impact actuator.

5.3 Performance

5.3.1 Sound and Haptic Stimuli. Fig. 13 shows examples of a visual scene, a sound signal, and a haptic stimulus generated from our conversion system. Persistent sound effects, such as car engines and shaking grounds, are classified as VIB. IMP usually occurs for events involving short and dull sounds; e.g., bashes with weapons and gunfires using a silencer. The sounds for MM and IMP are similar, but those for MM involve longer durations and larger reverberations; e.g., swords clashing and gunfires of a sniper rifle. The plots in Fig. 13 demonstrate that our system is capable of such selective conversions with short latencies.

5.3.2 Sound-to-Haptic Sensory Latency. We measured the sound-to-tactile sensory latency of our system. While playing an FPS game named Aim Lab [33], we fired the weapon ten times, and its sounds triggered our program running in parallel to make multimodal haptic stimuli. We recorded the sound stimuli using a microphone (C1, Mota Studio) attached to the wired speaker (Music Flow Portable Bluetooth Speaker, LG Electronics) and the haptic stimuli using an accelerometer (352C23, PCB Piezotronics) fastened to the mouse in Fig. 4. The signals were acquired and synchronized using a data acquisition card (PCIe-6323, National Instruments). We estimated the sensory latency between the sound and haptic signals by detecting the maximum changes of their RMS values, as described in Killick et al. [27]. The sensory latency from sound to impact was 63.3 ms on average (range: 54.6 and 73.9 ms). The sensory latency from sound to vibration was 22.9 ms on average (range: 8.8 and 33.9 ms). These latency values are smaller than or comparable to

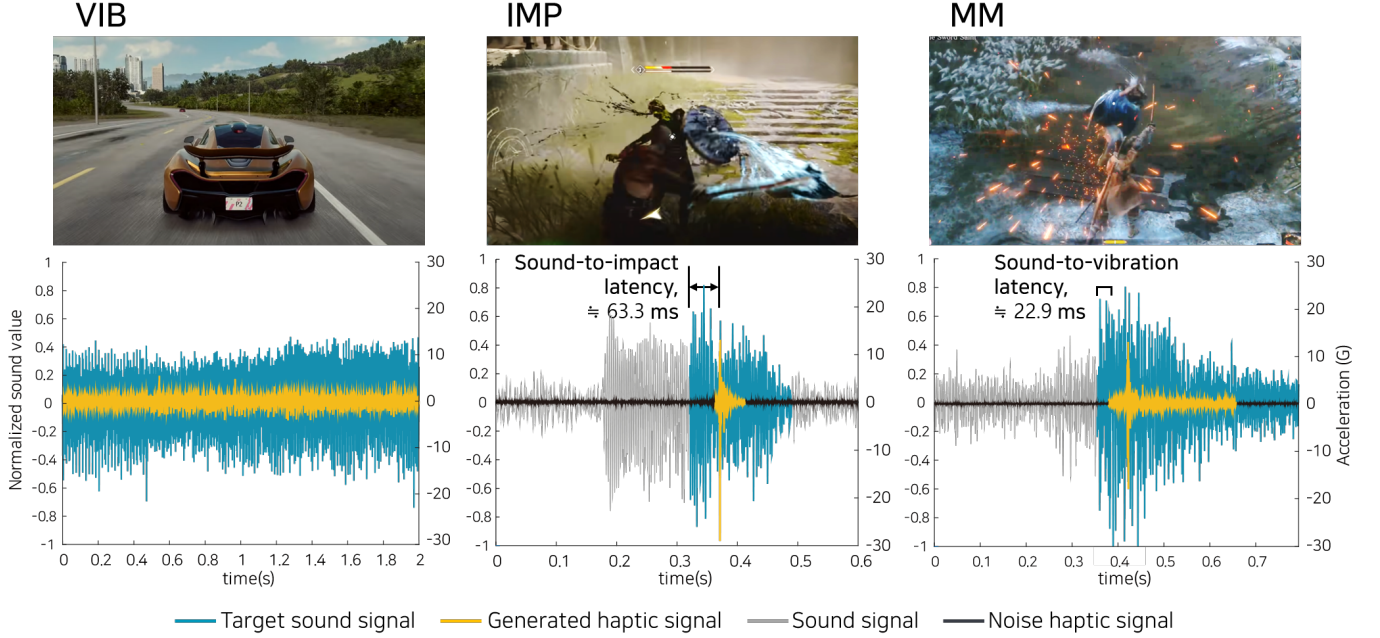


Figure 13: Visual scenes (top) and sound signals (bottom), and haptic signals generated by our system (bottom; overlayed), for VIB (car engine), IMP (blunt blow), and MM (two swords clashing) from left to right. The haptic signals were measured by an accelerometer attached to the customized mouse (Fig. 4).

the published gap detection thresholds between sound and tactile stimuli (20–75 ms [3, 45]). Therefore, our audio-to-tactile conversion system ensures synchronized multisensory experiences.

6 USER STUDY

The objective of this user study was to compare our sound-to-haptic conversion system with other methods in terms of user experiences. The independent variables were the conversion algorithm and the modality of haptic effect. This user study was approved by the Institutional Review Board at the leading authors' institution (PIRB-2022-R019).

6.1 Methods

6.1.1 Experimental Conditions. We considered the two scenarios of participants *actively* playing a game and *passively* watching a gameplay video. In the active mode, we observed that many participants were new to the selected games and focused on learning how to play the games instead of evaluating haptic effects. Hence, we decided to go with the passive mode, which has another important advantage that all participants receive identical visual-auditory-haptic stimuli. Although the user experiences in the active and passive gaming modes should be correlated to a certain extent, we acknowledge that they may not be completely the same.

We selected and downloaded two 90-s videos from YouTube for each of the two game genres. The videos for RPG were Sekiro (V-RPG-1) and Elden Ring (V-RPG-2). The video for FPS contained

two different segments from Call of Duty: Modern Warfare (V-FPS-1 and V-FPS-2). The game characters performed all their possible actions and skills in all the videos. The characteristics of the four videos are summarized in Table 2, and the full videos are available in the supplemental materials.

Two more real-time sound-to-tactile conversion algorithms were included in the experiments for comparison with the one described in this paper. One was a direct vibrotactile playback of sound signals after low-pass filtering (LPF-VIB). The LPF-VIB is the baseline method for demonstrating the effectiveness of selective conversion algorithms.

The other was the selective sound-to-vibration conversion algorithm using psychoacoustic features by Li et al. [38]⁶(PSY-VIB). This algorithm is based on the idea that sounds that are conducive to haptic effects are usually perceptually loud, sharp, deep, or intense in a low-frequency band as measured by loudness, sharpness, booming, and low-frequency energy, respectively. Each of the four measures has an individual threshold, and vibration feedback is presented when any measure exceeds its threshold. We used the thresholds recommended in [38] as follows: 0.85, 0.50, 0.65, and 0.80 for normalized loudness, sharpness, booming, and low-frequency energy, respectively. For normalization, we precomputed the maximum values of the measures from the videos used in the experiments. This psychoacoustics-based algorithm replaced the random forest sound-haptic classifier of our system, and only vibration stimuli were generated. This algorithm can be regarded as the SOTA.

⁶We greatly appreciate the authors of Li et al. [38] for sharing working codes with us.

Table 2: Characteristics of the four videos used in the user study.

Video	Sound Features	Source
V-RPG-1	Magnificent music, frequent clashes and swings of swords, occasional standoff attacks	5'00"–6'30" in [79]
V-RPG-2	Magnificent music, hits with a sword, occasional special moves, frequent standoff attacks	8'24"–9'54" in [80]
V-FPS-1	Frequent bass instruments, gunfire, occasional grenade explosions, conversations	6'13"–7'43" in [78]
V-FPS-2	Occasional bass instruments, gunfire, smashing a door, conversations	2:31'13"–2:32'43" in [77]

Our random forest-based sound-to-tactile conversion systems were used in two options: vibrations only (RF-VIB) or impact + vibration (RF-MM). This allowed us to check the effectiveness of multimodal haptic rendering. In RF-VIB, a vibration was generated even when the sound type was classified as IMP. As a result, participants compared the user experiences of multisensory stimuli provided by a total of four real-time sound-to-haptic conversion algorithms.

6.1.2 Procedure. The first step was confirmation of the discriminability between the vibration, impact, and multimodal stimuli. Every participant experienced the three stimuli and confirmed that s/he could easily distinguish them. The haptic stimuli were presented by the mouse (Fig. 4) grasped by the hand. Then, they proceeded to the next step.

We conducted two experiments, one for RPG and the other for FPS. In each experiment, participants conducted a practice and a main session for V1, and then V2 as prepared for the respective game genre. The order of the two videos was not balanced, as we did not expect it would affect the experiment results.

The practice sessions allowed participants to familiarize themselves with the testing environment and experimental procedure and to establish their own evaluation criteria for comparing different haptic effects. In each practice session, they viewed the first one minute of the assigned video while experiencing sound and haptic stimuli under one of the four haptic conditions (LPF-VIB, PSY-VIB, RF-VIB, and RF-MM). Then, they had a break over 30 s and proceeded to the next trial for a different haptic condition. The order of the haptic conditions was randomized per participant and video. After the practice sessions, we conducted an interview where participants described their experiences with each haptic condition to confirm the participants' understandings of the goals and procedures of the experiment.

In the main sessions, a full-length video was presented to participants along with the sound and haptic stimuli. After that, participants answered a questionnaire, followed by a 1-min break. Then, they proceeded to the next haptic condition. The orders of the four haptic conditions were balanced across participants using Latin squares.

The questionnaire consisted of six questions to evaluate user experiences of viewing gameplay videos with sound and haptic effects. The questions were:

Q1 (Adequacy) The haptic effects were adequate for the audiovisual context in the scenes.

Q2 (Distraction) The haptic effects distracted me while watching the video.

Q3 (Enjoyment) I enjoyed watching the video with the haptic effects.

Q4 (Synchrony) The haptic effects were synchronized in time with the audiovisual events.

Q5 (Fatigue) I felt tired after watching the video with the haptic effects.

Q6 (Preference) I liked watching the video with the haptic effects.

Participants indicated their degree of agreement to each question by continuously moving a slider on a straight line displayed on the monitor that represented a scale from 0 to 100. The line was marked at every 25 points: 0–strongly disagree, 25–disagree, 50–neutral, 75–agree, and 100–strongly agree. The Q2 (Distraction) and Q6 (Fatigue) scores were inverted during data processing so that a higher score indicates a better user experience in all metrics. After the scoring, participants were asked to share additional comments about the experimental condition.

6.1.3 Participants. Twenty volunteers took part in the RPG experiment (4 females and 16 males; 19–60 years old with a mean of 28.6) and another 20 for FPS sessions (11 females and 9 males; 19–50 years old with a mean of 27.5). None of them reported known sensorimotor disorders. The experiment took approximately one hour per participant. Each participant was paid approximately USD 13.5 after the experiment.

6.2 Results

Fig. 14a and 14b show the average scores of the six questions collected for the two game genres. A two-way repeated-measures ANOVA was applied with the two independent variables of *haptic condition* and *video*. The dependent variables were the six user-experience metrics. The statistical analyses were conducted independently for RPG and FPS.

Haptic condition was a significant factor for every measure in RPG [accuracy ($F(3, 57) = 7.86, p = 0.0002$); distraction ($F(3, 57) = 6.47, p = 0.0008$); enjoyment ($F(3, 57) = 3.22, p = 0.0292$); synchrony ($F(3, 57) = 4.05, p = 0.0111$); fatigue ($F(3, 57) = 16.84, p = 0 < 0.0001$); and satisfaction ($F(3, 69) = 4.19, p = 0.0095$)]. For FPS, *haptic condition* was also significant for accuracy [$F(3, 57) = 4.10, p = 0.0105$], distraction [$F(3, 57) = 4.81, p = 0.0047$], synchrony [$F(3, 57) = 8.63, p < 0.0001$], fatigue [$F(3, 57) = 10.55, p = 0 < 0.0001$], and satisfaction [$F(3, 69) = 3.00, p = 0.0380$], but not for enjoyment [$F(3, 57) = 2.42, p = 0.0757$]. *Video* was a significant factor for only adequacy in RPG [$F(1, 3) = 7.98, p = 0.0108$] and distraction in FPS [$F(1, 3) = 5.05, p = 0.0366$]. We also ran the Student-Newman-Keuls (SNK) post-hoc tests on the significant

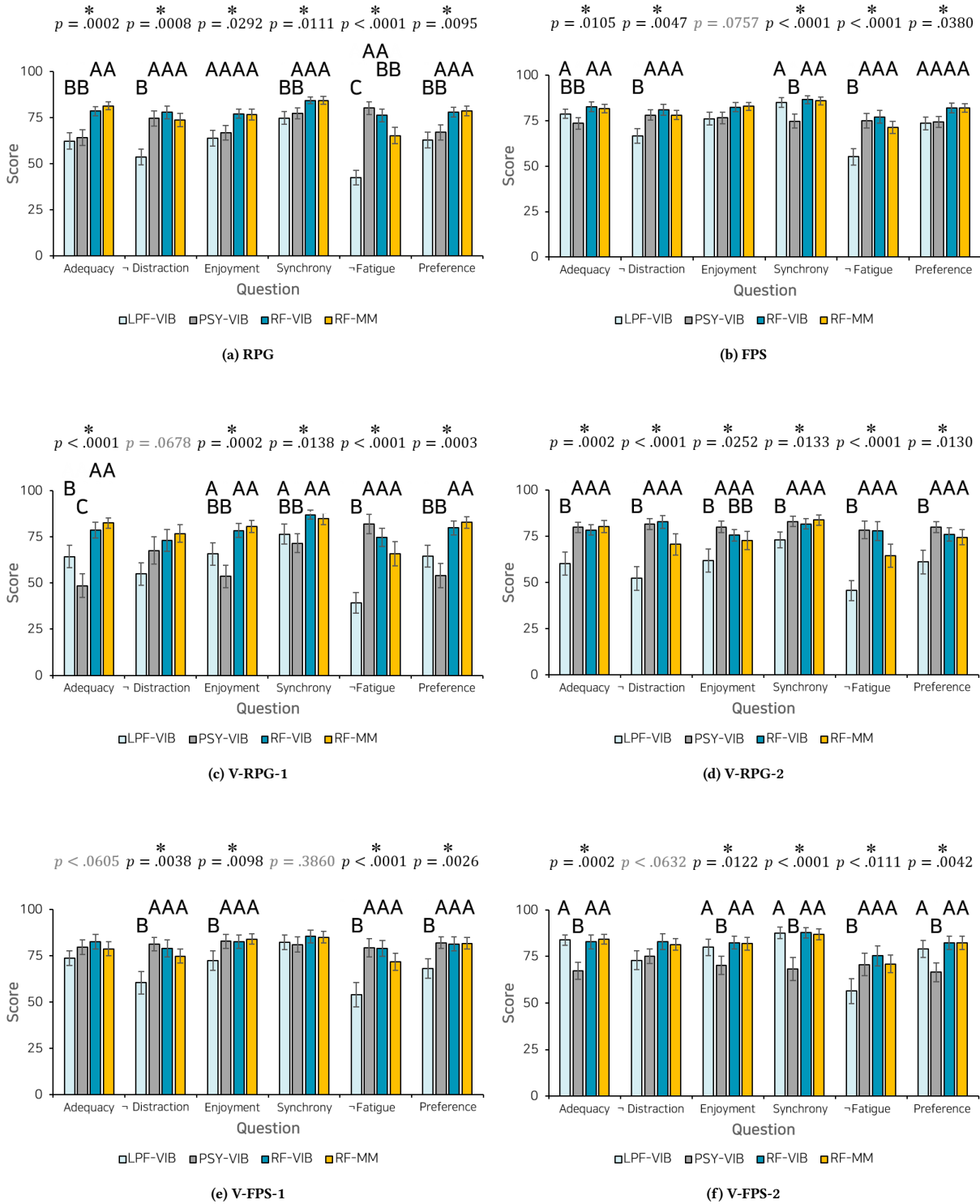


Figure 14: Results of the user study. Error bars represent standard errors. Significant effects ($p < 0.05$) are marked with asterisks. The haptic conditions grouped with the same letter indicate that they did not show significant differences by the SNK tests. Note that the distraction and fatigue scores are inverted, as denoted by a $\neg$ sign.

haptic conditions. The average scores and the grouping results are represented above the bars in Fig. 14a and 14b.

Significant interaction terms between *haptic condition* and *video* were found for all measures, except fatigue in RPG and distraction in both RPG and FPS (see the supplemental material for all the results). To investigate the interaction effects further, we conducted one-way repeated-measures ANOVAs for each video with *haptic condition* as the independent variable. Then, the SNK test was applied to the significant cases. Fig. 14c–14f show the average scores of the six measures and the grouping results by the SNK tests.

6.3 Discussion

The average scores of RF-MM were 81.4 in adequacy, 79.7 in enjoyment, 85.1 in synchrony, and 80.3 in preference along with 24.1 in distraction and 31.8 in fatigue, all on a scale of 0–100. As users generally avoid extreme scores, these results provide solid evidence for the effectiveness of our automatic sound-to-haptic conversion system. In what follows, we compare the four haptic conditions in detail to reveal deeper and more useful insights.

6.3.1 Effects of Selective Conversion. Comparing PSY-VIB and RF-VIB with LPF-VIB allows us to assess the effectiveness of the selective sound-to-haptic conversion algorithms over the classical direct sound playback approach. For RPG, RF-VIB scored significantly better than LPF-VIB in all measures except enjoyment (Fig. 14a). PSY-VIB showed significantly better (i.e., lower) scores than LPF-VIB in distraction and fatigue. These results demonstrate that the selective conversion algorithm provides improved user experiences than the direct sound playback for RPG games.

RPG games generally use loud BGM to make gameplay immersive and induce gamers into certain moods. LPF-VIB converts all of these sounds to haptic effects. In fact, concentrating on the gameplay scenes was difficult with LPF-VIB (15 Ps⁷ out of 20) due to the constant vibrations. The haptic effects generated by LPF-VIB felt less correlated with the audiovisual content, thereby hampering user experiences (3 Ps). Also, the continuous and long vibrotactile effects presented by LPF-VIB felt tiresome (5 Ps).

It is noteworthy that LPF-VIB resulted in a significantly lower score in synchrony than RF-VIB. This is somewhat unexpected because by design LPF-VIB provides vibration effects with no time delay. A potential reason could be that the continuous haptic effects generated from the strong BGM masked the perception of haptic effects for other important events, and as a result they lowered the sense of synchrony.

For V-RPG-2 (Fig. 14d), RF-VIB and PSY-VIB showed significantly better user experiences than LPF-VIB in all measures except enjoyment, similar to the average results. V-RPG-2 accompanied strong BGM, and it disabled the generation of appropriate haptic effects (4 Ps).

For V-RPG-1 (Fig. 14c), LPF-VIB elicited somewhat better user experiences. In particular, LPF-VIB was significantly better in adequacy than PSY-VIB even though LPF-VIB often incurred inadequate vibration effects generated from non-target sounds. PSY-VIB

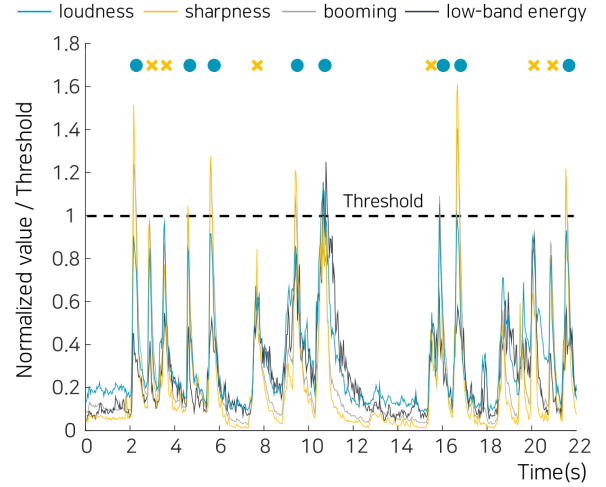


Figure 15: Psychoacoustic feature values of PSY-VIB computed from V-RPG-1. The vertical axis indicates the normalized feature values divided by each threshold. Circles above the plot mean vibrations are generated at that moment by both RF-VIB and PSY-VIB from the same events, while X marks are only by RF-VIB.

frequently failed to detect important sound events and did not generate haptic effects for this video. It was because the simple thresholding scheme of psychoacoustic features used for detection was incapable of handling the various distributions of the PSY-VIB feature values. Fig. 15 represents such a phenomenon in V-RPG-1. The thresholds increased with the prominent maximum feature values, prohibiting the several target sounds from being converted into vibration. The true-negative rate of PSY-VIB was almost 43% in 22 s, while RF-VIB successfully detected all the target sounds. It appears that the participants preferred the more frequent vibration effects generated by LPF-VIB, although some of which were unnecessary, to the more scarce haptic effects of PSY-VIB caused by detection misses (5 Ps).

As for the FPS games, the average scores of LPF-VIB over the two videos did not differ significantly from those of PSY-VIB and RF-VIB, except for fatigue and distraction (Fig. 14b). LPF-VIB even showed a higher synchrony score than PSY-VIB. FPS games also contain sounds unsuitable for haptic feedback, such as conversation and music. Here, the conversation sounds usually have low energy within the bandwidth of vibration conversion (under 250 Hz), and they do not seem to cause critical problems to LPF-VIB. Also, FPS games use BGM much more sparingly, as being able to hear enemies' positions is important. Without the haptic feedback associated with conversations and BGM, LPF-VIB can be comparable to selective sound-to-tactile conversion, at least in terms of user experiences measured in our experiments.

In the video V-FPS-1 (Fig. 14e), LPF-VIB showed significantly lower scores in distraction, enjoyment, fatigue, and preference than PSY-VIB and RF-VIB. In this video, BGM was relatively loud during the battles, and the haptic effects of gunfire were not distinguishable from those corresponding to BGM (6 Ps). In contrast, in V-FPS-2 (Fig. 14f), LPF-VIB had a significantly lower score than RF-VIB

⁷This notation means that the preceding statement was reported by 15 participants.

in only fatigue. PSY-VIB received significantly lower scores than LPF-VIB in adequacy, enjoyment, synchrony, and preference. The reason was that PSY-VIB often missed haptic conversion of important sound events, which was regarded as inconsistency (9 Ps).

Direct sound playback is a very efficient rendering method of haptic effects. However, LPF-VIB always scored the worst (i.e., highest) in distraction and fatigue. Participants' hands felt numb after experiencing LPF-VIB (4 Ps RPG and 7 Ps FPS). Playing a game with LPF-VIB for a long time also seemed difficult (5 Ps RPG), presumably due to the excessive vibrations.

6.3.2 Effects of ML-based Classification. The effects of the sound classification method can be understood by comparing the experimental results between our ML-based algorithm RF-VIB and the current SOTA, PSY-VIB. As shown in Fig. 14a, RF-VIB demonstrated better average scores than PSY-VIB in every measure except for fatigue, but the differences were significant only for adequacy. This is an example where the higher classification accuracy of the RF-based event selection contributed to better user experiences.

PSY-VIB showed quite different results in the two RPG videos. In V-RPG-1, RF-VIB outperformed PSY-VIB in adequacy, enjoyment, synchrony, and preference (Fig. 14c). As mentioned earlier, PSY-VIB was not very tailored to V-RPG-1. However, in V-RPG-2, PSY-VIB and RF-VIB provided similar levels of user experiences (Fig. 14d). In this video, the levels of the psychoacoustic features were relatively well separated between the target sounds and the BGM. Therefore, PSY-VIB, the thresholding algorithm, could provide similar haptic effects to those of RF-VIB. Some participants noticed no differences between PSY-VIB and RF-VIB (6 Ps).

On average, the user experiences on the FPS videos were not clearly different between PSY-VIB and RF-VIB with no significant differences in any measures (Fig. 14b). The same was observed in the results of V-FPS-1 (Fig. 14e), where no differences between PSY-VIB and RF-VIB were reported (3 Ps). Contrarily, in V-FPS-2, the scores in adequacy, enjoyment, synchrony, and preference were significantly better for RF-VIB than PSY-VIB (Fig. 14f). This seems to be due to the same reason of the inadequately high thresholds used by PSY-VIB.

The sound-to-haptic conversion quality of PSY-VIB is greatly influenced by the sound characteristics and the predetermined thresholds. Our ML-based algorithms are relatively free from such requirements and can offer high-quality haptic conversion evenly.

6.3.3 Effects of Multimodal Haptic Effects. In this study, no significant differences appeared between RF-MM and RF-VIB in any measures. For a deeper understanding, we analyzed the interviews and comments of the participants. It does not appear that this result was not due to the participants' inability to distinguish between the two haptic conditions. Rather, there was a large variance in individual preferences. For example, the participants who preferred RF-MM considered the impact stimulus as a realistic sensation for collisions and liked the immersive and novel experiences that it provided (8 Ps RPG and 8 Ps FPS). To the contrary, some participants preferred RF-VIB because they found the strong impact stimuli to be excessive or tiring (4 Ps RPG and 3 Ps FPS). In particular, the participants who gave RF-MM lower scores than RF-VIB in RPG commented that the impact stimuli were not harmonious with the sounds for ranged-attacks occurring without collisions

(3 Ps). Regarding the FPS games, using the impulse stimuli at a constant (maximum) intensity for different situations (e.g., gun firing sounds of the main character and other unimportant characters) was not suitable. For further improvements, it seems necessary to improve upon both the function of the impact actuator and its rendering algorithm.

7 CURRENT LIMITATIONS AND FUTURE WORK

Our sound-to-haptic conversion system based on machine learning generates multimodal haptic effects in real time and provides significantly better user experiences compared to the SOTA algorithms for the same purposes, as demonstrated in the present study. Our system still has a few limitations that warrant future work.

7.1 Sound Classification

Given a sound sample, our system determines the type of haptic stimulus to accompany it, but it does not classify the sound class itself; e.g., whether the sound originates from a gunshot, an explosion, or something else. If the sound class is identified, we can design highly tailored haptic effects. For example, for explosion sounds, we can add low-frequency rumbling components to the haptic effects. For gunfire sounds, we can even elicit a haptic illusion that feels like a vibration penetrating the body using haptic suits [28]. Sound class identification can also help with the detection of multiple sound events within a short window length (40 ms). This information is useful for rendering a single long haptic effect versus multiple short effects. We expect that the sound-based event classification will expand the design space of haptic effects and provide additional user benefits. To this end, we may need to try other effective and efficient ML models, such as those using sequential models widely used for sound classification.

For broader use, a universal sound-to-haptic classifier is ideal that deals with various sound classes and are robust to the types and genres of multimedia content. For example, when tested, our classifier built from the RPG dataset performed well for FPS games, and vice versa. It can be because impulsive sounds, our primary target, have common temporal or spectral features independent of the semantics of sounds compared to the other types of sounds. This advantage can be a foundation for developing universal classifiers for haptic feedback purposes.

Another issue, which is more subtle, is related to the way we built the sound-haptic datasets. In this work, the haptic experts labeled the appropriate type of haptic feedback while listening to sounds only, without watching the visual context. The participants in the user study were presented with a few examples of inadequate haptic effects; e.g., haptic effects generated for gun firing sounds of other unimportant characters and impact effects for magic-like long ranged shots. Such events, if the visual contexts are given, may be labeled as other haptic types such as NONE and VIB, respectively, for the two preceding examples. If such events involve unique characteristics to the sounds, such as differences in left-right balance or phase in stereo signals, then they might be learned by an ML classifier for improved classification performance.

7.2 Multimodal Haptic Rendering

We also need to improve the impact and vibration rendering methods used in our conversion system. At the moment, our system always generates an impact stimulus at the maximum intensity and waits for 40 ms to enable the next impact. This simplified approach may be responsible for the lack of apparent differences between RF-VIB and RF-MM in the user experiences (Section 6.2). Also, we drive a vibration actuator with a sound signal after simple low-pass filtering. The resulting haptic effect responds to all the low-frequency components in the sound, including BGM, regardless of their adequacy. An algorithm that separates event and background sounds and renders only the target sound may improve the rendering quality further.

The core algorithms of our work may be extended to other forms of multimodal haptic systems. For instance, the impact stimuli can be replaced with kinesthetic feedback as in Danieau et al. [14] or vestibular motion feedback as in Yun et al. [81]. Vibration feedback can be enriched by using full-body haptic suits, which have recently been commercialized [6]. Such extensions could be a fruitful research direction to provide more immersive and realistic experiences to users.

7.3 User Experience Evaluation

We adopted passive gameplay to evaluate user experiences in this work. As mentioned earlier, the main reason was the participants' difficulty in evaluating haptic experiences when they focused on playing unfamiliar games. An effective strategy in this situation can be a longitudinal user study, which allows us to observe the changes and trends in user experiences as participants become more experienced over a sufficiently long period. Comparing the results between the passive and active modes may shed new insights on the effects of haptic feedback on enhancing gaming experiences.

7.4 Usage in Haptic Authoring Pipeline

Automatic classification of the sound events suitable for haptic emphasis, demonstrated in our work, is also useful for manual authoring of haptic effects for multimedia content. Such a function can annotate the sound events with appropriate haptic feedback type, thereby alleviating the designers' efforts for segmenting the audiovisual stream. The detected sound segments can be processed by adequate haptic rendering algorithms to compute haptic effect commands. Then, these commands can be refined manually by the designers. These constitute another automated function for saving the authoring cost. Integrating such automated algorithms into existing manual authoring tools is a promising direction for wider dissemination of the multisensory applications and content exploiting haptic effects.

8 CONCLUSIONS

To enhance interactive and real-time gameplay experiences, we developed a system that provides haptic effects consisting of vibration and impact stimuli using only sound signals. The system runs in real time, generating haptic effects synchronized with the audio stream. To this end, our system determines whether the sound

is appropriate for an accompanying haptic effect. If so, it also selects the type of multimodal haptic stimulus in the form of a vibration, an impact, or a vibration+impact effect using an ML-based approach. Our sound type classifier performs at a high classification accuracy; 98.7% tested with our augmented datasets. The system then produces the haptic stimulus based on the selected type. A user study validated that our system elicits better user experiences compared to the current SOTA algorithms for the same crossmodal conversion purposes. The limitations of our current system and directions for future work are also discussed.

ACKNOWLEDGMENTS

This work was supported in part by a research award from Google entitled "Semantic Sound-to-Haptic Conversion: Algorithms and Performance Metrics," Samsung Research Funding & Incubation Center (No. SRFC-IT1802-05), and by the Mid-Career Researcher Program of the National Research Foundation of Korea (No. 2022R1A2C2091161).

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